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QUALIFICATION REQUIREMENTS REMOVED

MIL-M-38510/45B
5 April 1984
SUPERSEDING
MIL-M-38510/45A
24 May 1982

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, TTL, LOW POWER,

BISTABLE LATCHES, MONOLITHIC SILICON

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, low power, bistable latch microcircuits. One product assurance class and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, with the exception that the "JAN" or "J" certification shall not be used.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	4-Bit latch, master reset
02	Dual, 4-bit latch

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline, (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
J	D-3 (24-lead, 1/2" x 1 1/4"), dual-in-line package
K	F-6 (24-lead, 3/8" x 5/8"), flat package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 to 7.0 V dc
Input voltage range	-1.5 V dc at -10 mA to 5.5 V dc
Storage temperature	-65° to +150°C
Maximum power dissipation, (P_D) 1/	
Device type 01	175 mW
Device type 02	340 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction to case (θ_{JC}):	
Cases E and J	0.08°C/mW
Cases F and K	0.09°C/mW
Junction temperature (T_J)	+175°C

1/ Must withstand the added P_D due to short circuit tests (e.g., I_{OS}).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center, (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5962

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V minimum to 5.5 V maximum
Minimum high level input voltage (V_{IH})	2.0 V
Maximum low level input voltage (V_{IL})	0.7 V
Normalized fanout (each output) 1/	5 maximum
Case operating temperature range (T_C)	-55°C to +125°C
Setup time, $t_{(SETUP)}$	
Device type 01:	
Data to enable (high)	10 ns, minimum
Data to enable (low)	20 ns, minimum
Data to set (high)	15 ns, minimum
Device type 02:	
Data to enable (high)	8 ns, minimum
Data to enable (low)	18 ns, minimum
Input hold time, $t_{(HOLD)}$	
Device type 01:	
Data to enable (high)	0 ns, maximum
Data to enable (low)	5 ns, maximum
Data to set (low)	0 ns, maximum
Device type 02:	
Data to enable (high)	0 ns, maximum
Data to enable (low)	4 ns, maximum

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specifictaion to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Case outlines. The case outlines shall be as specified in MIL-M-38510 and in 1.2.3 herein.

3.2.2 Terminal connections. The terminal connections shall be as specified on figure 1.

1/ Device will fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
High level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -400 µA, V_{IN} = 0.7 V	01,02	2.4	---	V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 4.8 mA, V_{IN} = 2.0 V	01,02	---	0.3	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, V_{IN} = -10 mA	01, 02	---	-1.5	V
Low level input current, data	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.3 V	01,02	-282	-640	µA
Low level input current, $\overline{S}$, $\overline{E}$, and $\overline{MR}$	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.3 V	01,02	-176	-400	µA
High level input current, data	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V 1/	01,02	0	30	µA
High level input current, data	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V 1/	01,02	---	150	µA
High level input current, $\overline{S}$, $\overline{E}$, and $\overline{MR}$	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 2.4 V 1/	01,02	---	20	µA
High level input current, $\overline{S}$, $\overline{E}$, and $\overline{MR}$	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V 1/	01,02	---	100	µA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V 2/ 3/	01,02	-2.5	-25	mA
Supply current	I_{CC}	V_{CC} = 5.5 V 2/; V_{IN} = 4.5 V	01	---	16.5	mA
			02	---	29	mA
To high level, from enable to output	t_{PLH1}	C_L = 50 pF; V_{CC} = 5.0 V; R_L = 1.78 kΩ ±5% for device type 01; R_L = 1.6 kΩ ±5% for device type 02	01	15	65	ns
			02	18	80	
To low level, from enable to output	t_{PHL1}		01	15	57	ns
			02	20	80	
To high level, from data to output	t_{PLH2}		01	8	50	ns
			02	15	42	
To low level, from data to output	t_{PHL2}		01	8	50	ns
			02	18	60	
To low level, from master reset to output	t_{PHL3}		01	8	50	ns
			02	15	60	
To high level, from set to output	t_{PLH4}		01	12	53	ns

1/ All unspecified inputs at 5.5 volts.

2/ All unspecified inputs grounded.

3/ Not more than one output should be shorted at a time.

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3.2.3 Truth tables. The truth tables shall be as specified on figure 2.

3.2.4 Logic diagrams. The logic diagrams shall be as specified on figure 3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. Electrical test requirements shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)
	Class B devices
Interim electrical test parameters (pre burn-in) (method 5004)	1
Final electrical test parameters (method 5004)	1*,2,3,7, 9
Group A test requirements (method 5005)	1,2,3,7, 9,10,11
Group C end-point electrical parameters (method 5005)	1,2,3
Group D end-point electrical parameters (method 5005)	1,2,3
Additional electrical subgroups for group C periodic inspections	None

*PDA applies to subgroup 1 (see 4.2c).

3.6 Marking. Marking shall be in accordance with MIL-M-38510. The "JAN" or "J" certification mark shall not be used.

3.7 Manufacturer eligibility. To be eligible to supply microcircuits to this specification, a manufacturer shall have manufacturer certification in accordance with MIL-M-38510 for at least one line. Not necessarily the line producing the device type described herein.

3.8 Certification. Certification in accordance with MIL-M-38510 is not required for this device.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

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4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection is not required.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4). Generic test data (see 6.7) may be used to satisfy the requirements for groups C and D inspections. Quality conformance inspection shall be completed on the specific devices covered by this specification before they are shipped.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, 6, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

c. Steady state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

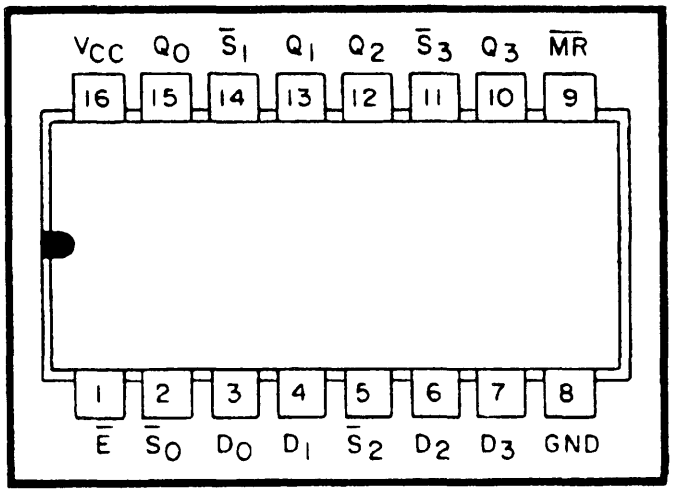
(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510.

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Device type 01

Cases E and F



Device type 02

Cases J and K

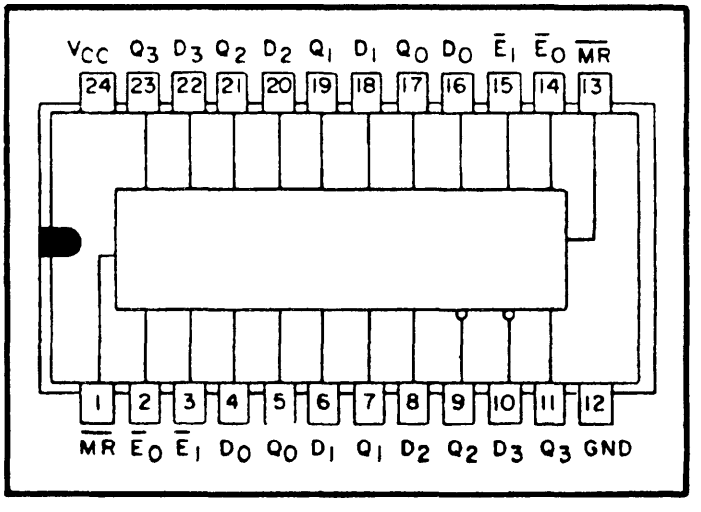


FIGURE 1. Terminal connections.

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Device type 01

Truth table					
$\overline{MR}$	$\overline{E}$	D	$\overline{S}$	Q_{n-1}	Operation
H	L	L	L	L	D mode
H	L	H	L	H	
H	H	X	X	Q_{n-1}	
H	L	L	L	L	R/S mode
H	L	H	L	H	
H	L	L	H	L	
H	L	H	H	Q_{n-1}	
H	H	X	X	Q_{n-1}	
L	X	X	X	L	Reset

Device type 02

Truth table					
$\overline{MR}$	$\overline{E_0}$	$\overline{E_1}$	D	Q_n	Operation
H	L	L	L	L	Data entry
H	L	L	H	H	Data entry
H	L	H	X	Q_{n-1}	Hold
H	H	L	X	Q_{n-1}	Hold
H	H	H	X	Q_{n-1}	Hold
L	X	X	X	L	Reset

NOTES:
X = Don't care
L = Low voltage level
H = High voltage level
Q_{n-1} = Previous output state
Q_n = Present output state

FIGURE 2. Truth tables.

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Device type 01

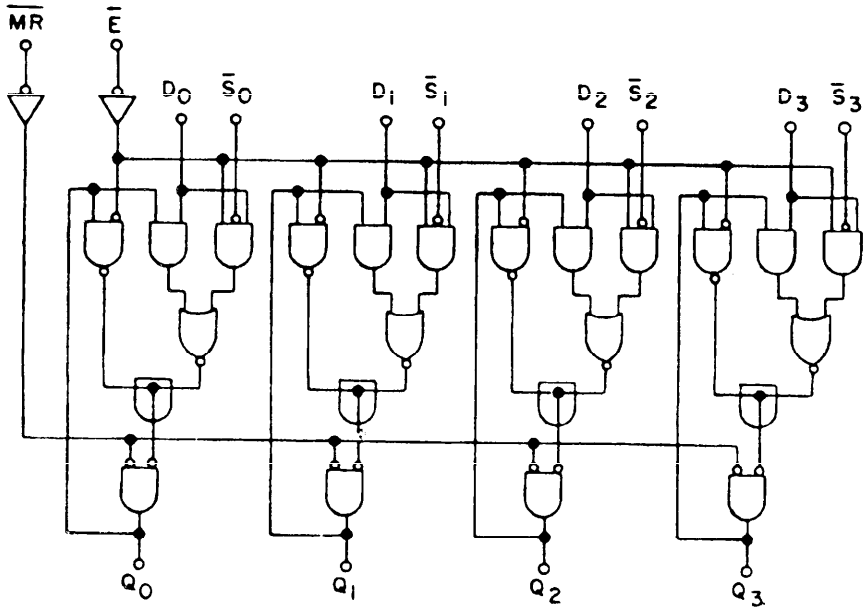
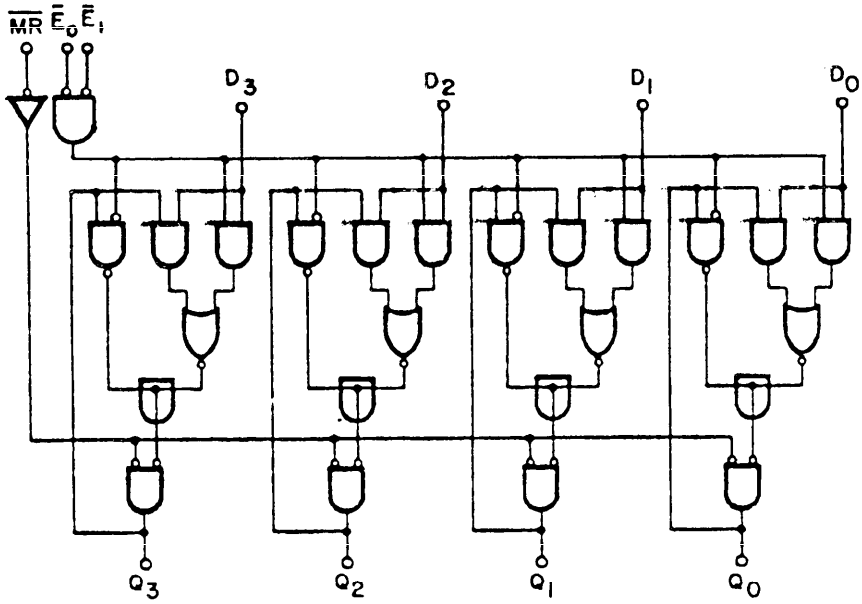


FIGURE 3. Logic diagrams.

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Device type 02

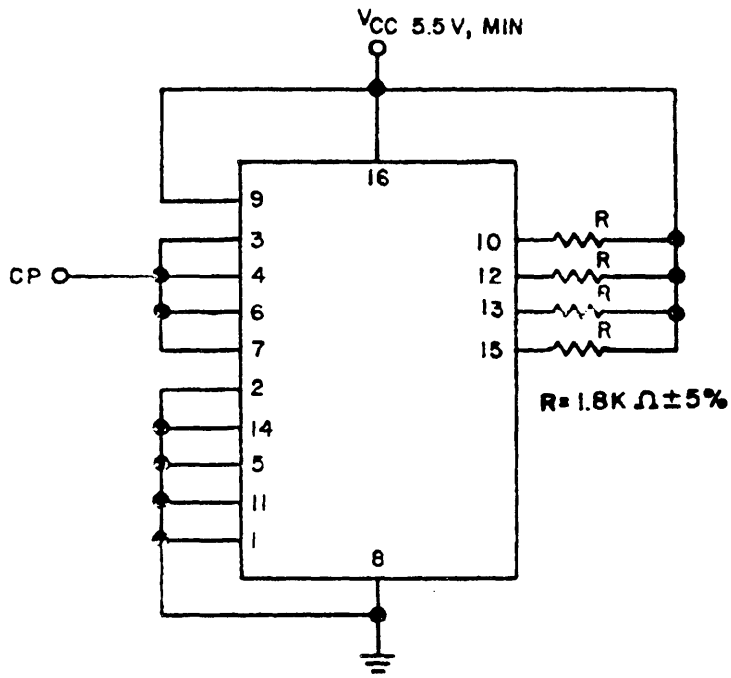


NOTE: Only one latch shown.

FIGURE 3. Logic diagrams - Continued.

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Device type 01

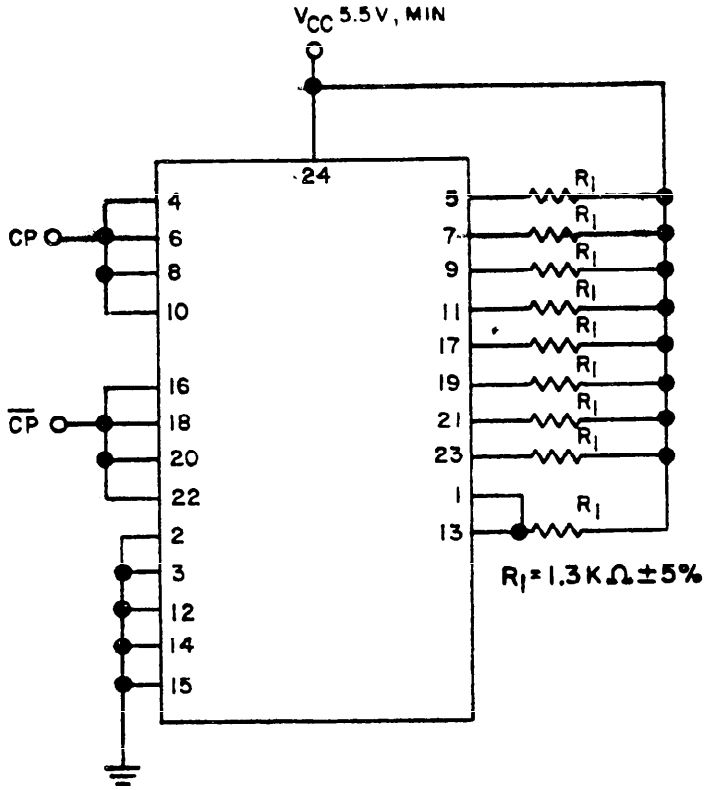


NOTE:
CP = 1 MHz ±50% square wave; duty cycle = 50 ±15%; V_{IH} = 2.0 V to 5.5 V; V_{IL} = -0.5 V to 0.7 V.

FIGURE 4. Burn-in and life test circuit.

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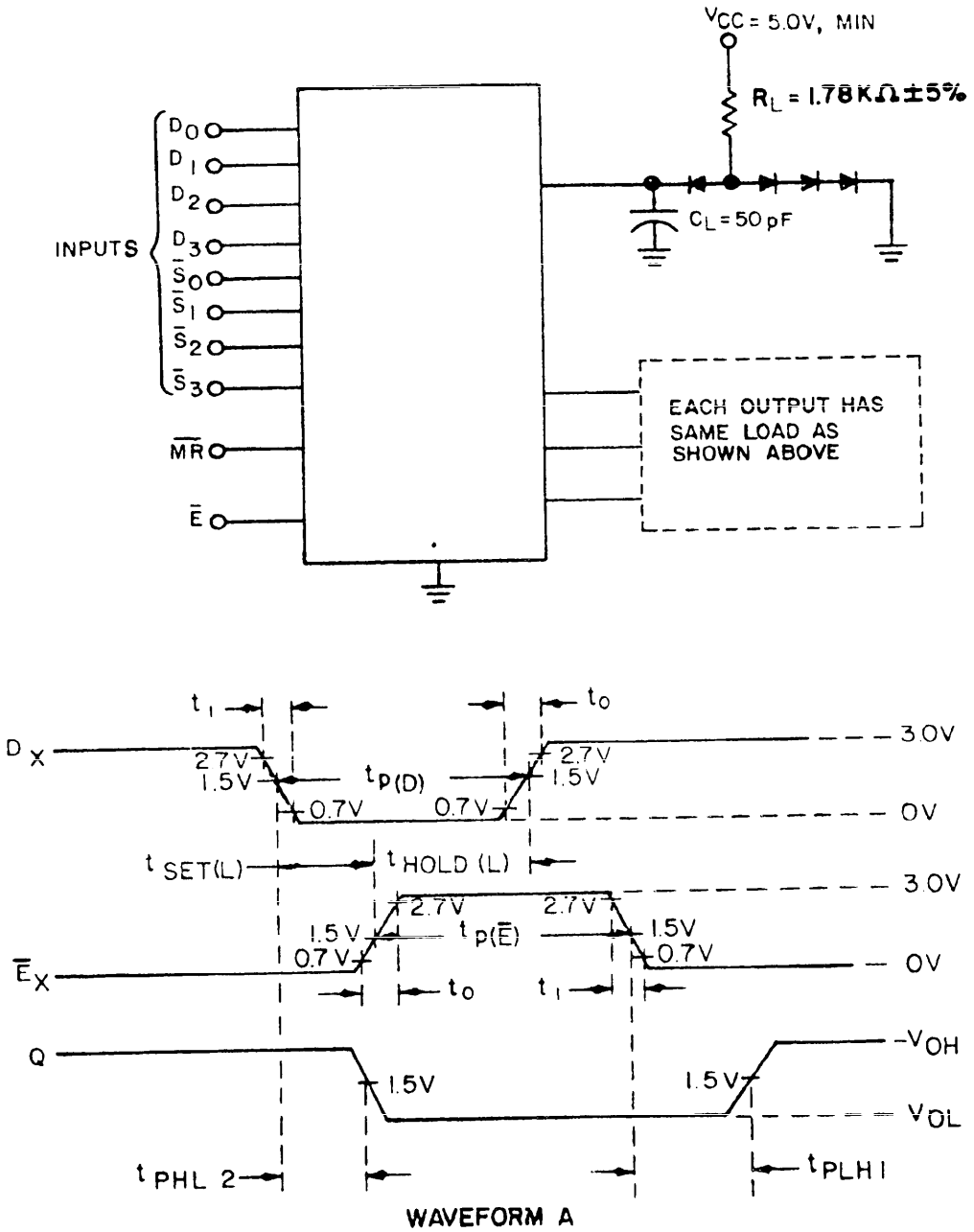
Device type 02



NOTE:
CP and $\overline{CP}$ = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IH} = 2.0 V to 5.5 V; V_{IL} = -0.5 V to 0.7 V.

FIGURE 4. Burn-in and life test circuit - Continued.

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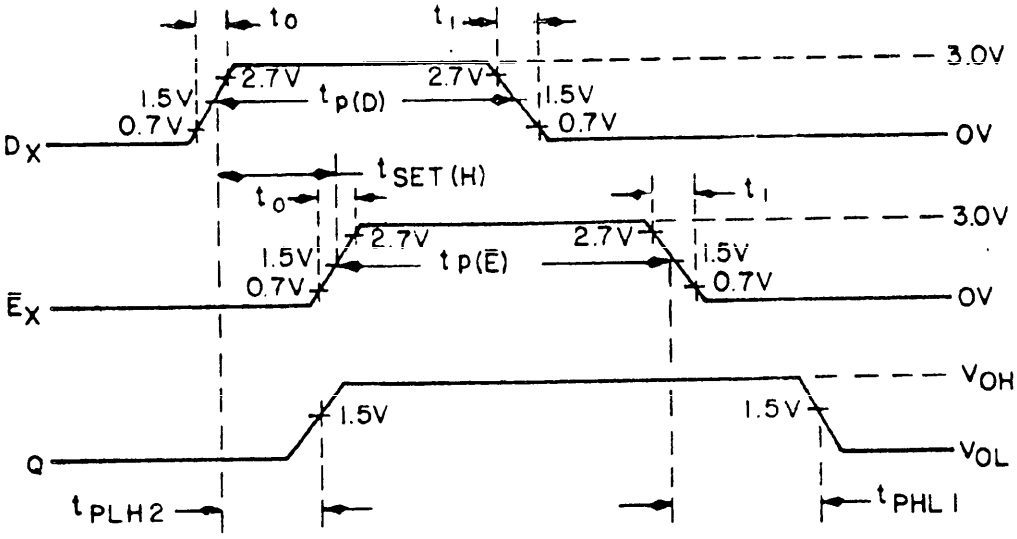


WAVEFORM A

NOTES:
1. The data inputs have the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{p(D)}$ = 32 ns, and PRR = 1 MHz.
2. The enable input has the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 10$ ns $t_{p(\overline{E})}$ = 20 ns, $t_{set(L)}$ = 25 ns, $t_{HOLD(L)}$ = 7 ns, PRR = 1 MHz and $Z_{out} \approx 50\ \Omega$.
3. Each latch is tested separately.
4. C_L = 50 pF, which includes probe and jig capacitance.
5. R_L = 1.78 kΩ ±5%.
6. All diodes are 1N3064 or equivalent.

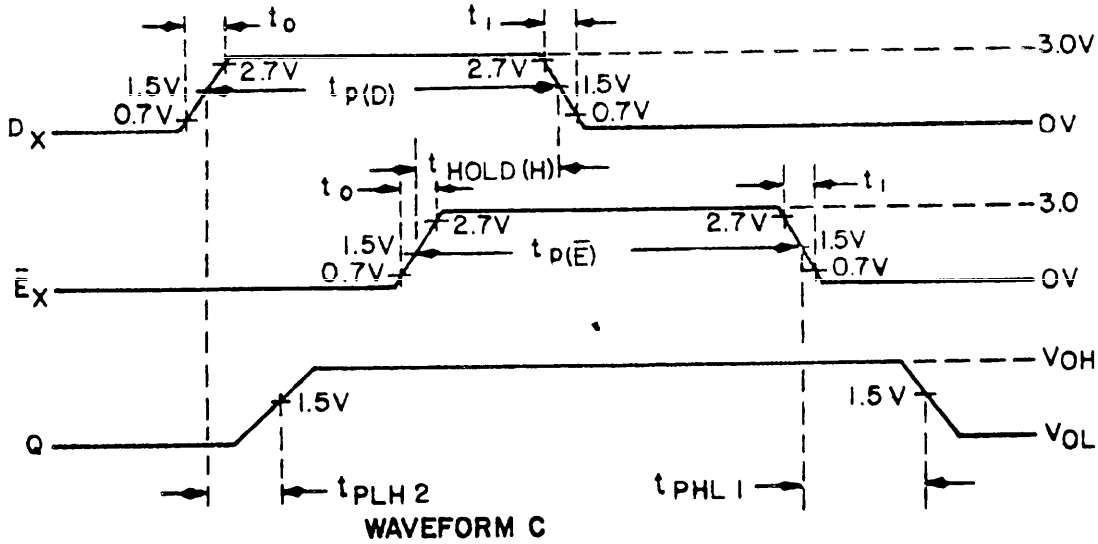
FIGURE 5. Switching time test circuit and waveforms for device type 01.

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WAVEFORM B

NOTES:
1. The data inputs have the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{p(D)}$ = 20 ns and PRR = 1 MHz.
2. The enable input has the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{p(\bar{E})}$ = 20 ns, $t_{SET(H)}$ = 5 ns, PRR = 1 MHz and $Z_{out} \approx 50\ \Omega$.

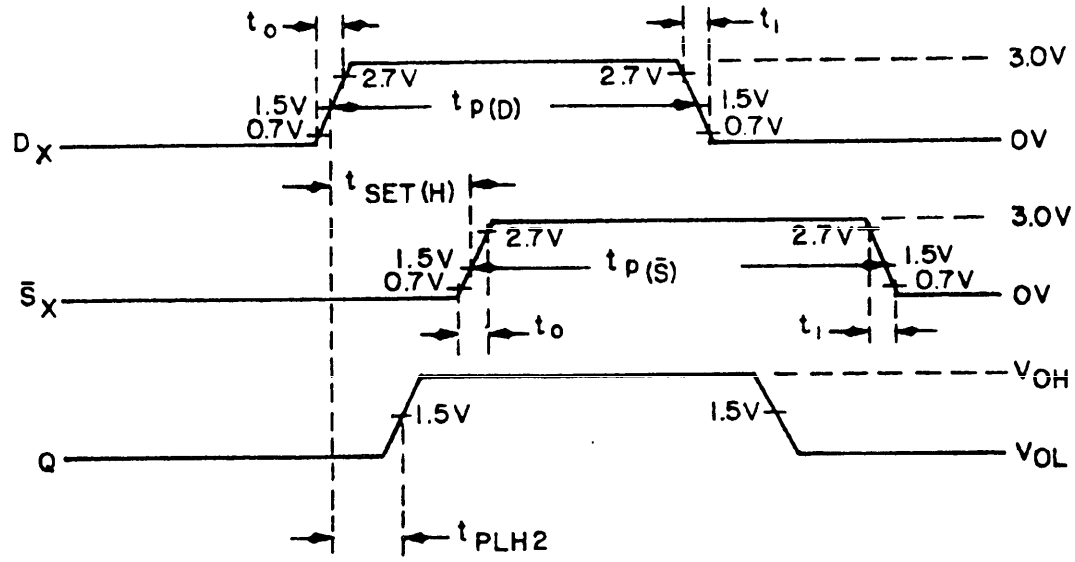


WAVEFORM C

NOTES:
1. The data inputs have the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{p(D)}$ = 20 ns and PRR = 1 MHz.
2. The enable input has the following characteristics: Vgen = 3 V, t0 = $t_1 \leq 10$ ns, $t_{p(\bar{E})}$ = 20 ns, $t_{HOLD(H)}$ = 0 ns, PRR = 1 MHz and $Z_{out} \approx 50\ \Omega$.

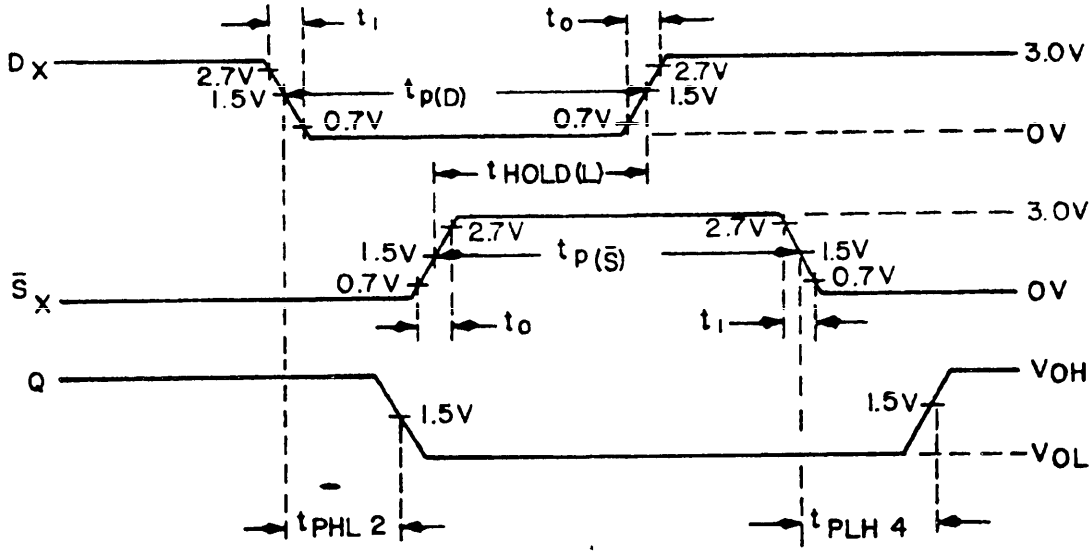
FIGURE 5. Switching time test circuit and waveforms for device type 01 - Continued.

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WAVEFORM D

NOTES:
1. The data inputs have the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{p(D)}$ = 20 ns, and PRR = 1 MHz.
2. The set inputs have the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{p(\bar{S})}$ = 20 ns, $t_{SET(H)}$ = 8 ns, PRR = 1 MHz and $Z_{out} \approx 50\ \Omega$.

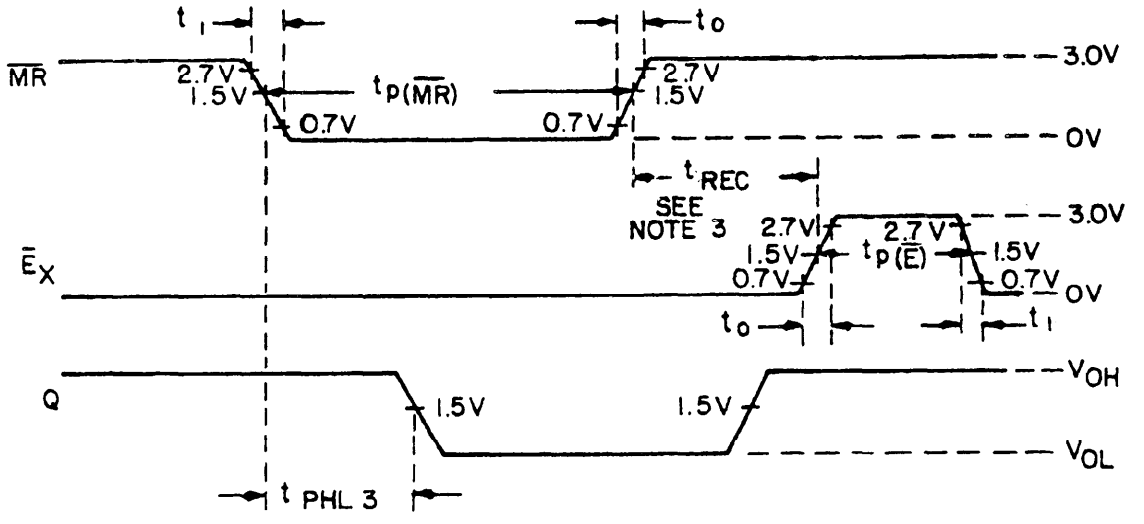


WAVEFORM E

NOTES:
1. The data inputs have the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{p(D)}$ = 20 ns, and PRR = 1 MHz.
2. The set inputs have the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{p(\bar{S})}$ = 20 ns, $t_{HOLD(L)}$ = 8 ns, PRR = 1 MHz and $Z_{out} \approx 50\ \Omega$.

FIGURE 5. Switching time test circuit and waveforms for device type 01 - Continued.

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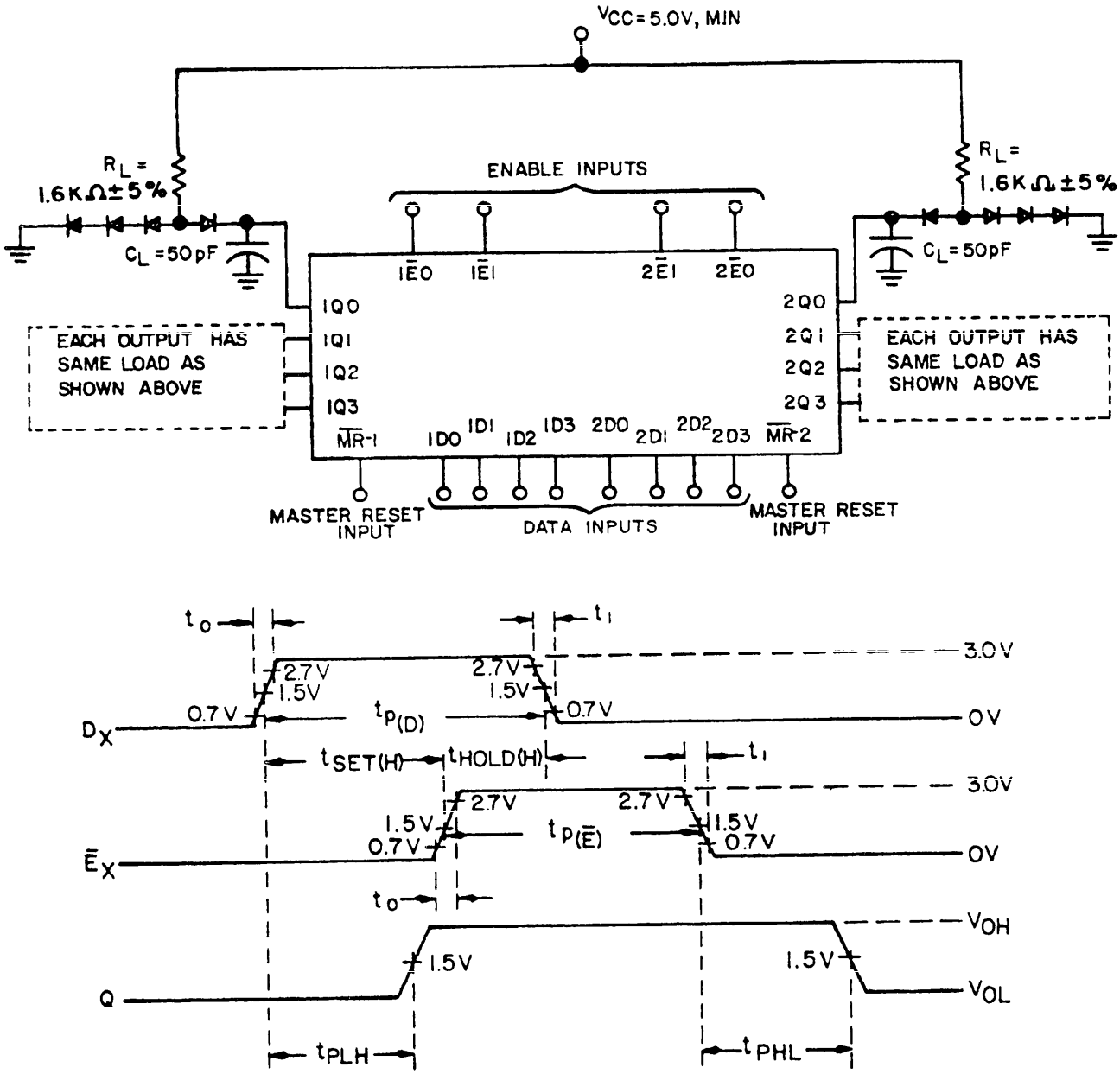


WAVEFORM F

NOTES:
1. The master reset input has the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{p(\overline{MR})}$ = 18 ns, and PRR 1 MHz.
2. The enable input has the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{p(\overline{E})}$ 20 ns, t_{REC} = 0 ns, PRR = 1 MHz, and $Z_{out} \approx 50\ \Omega$.
3. Recovery time is the minimum time that enable must remain low after the master reset transition from low to high in order for the latch to recognize and store high data.

FIGURE 5. Switching time test circuit and waveforms for device type 01 - Continued.

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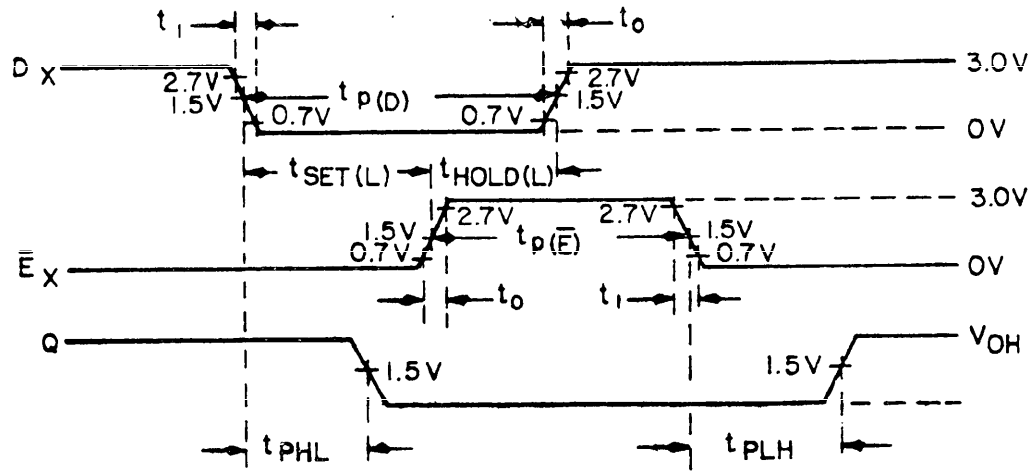
WAVEFORM A

NOTES:

1. The data inputs have the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{p(D)}$ = 16 ns, and PRR = 1 MHz.
2. The enable inputs have the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq$ 10 ns, $t_{SET(H)}$ = 10 ns, $t_{HOLD(H)}$ = 0 ns, $t_{p(E)}$ = 20 ns, PRR = 1 MHz and $Z_{out} \approx 50\ \Omega$.
3. Each latch is tested separately.
4. C_L = 50 pF, which includes probe and jig capacitance.
5. R_L = 1.6 kΩ ±5%.
6. All diodes are 1N3064 or equivalent.

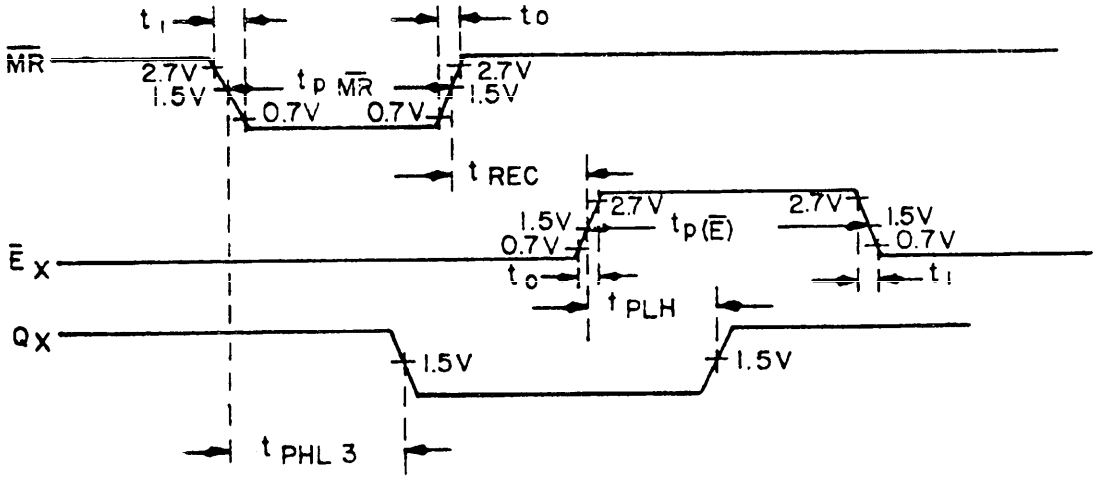
FIGURE 5. Switching time test circuit and waveforms for device type 02.

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WAVEFORM B

NOTES:
1. The data inputs have the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{p(D)}$ = 26 ns, and PRR = 1 MHz.
2. The enable inputs have the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{SET(L)}$ = 16, $t_{HOLD(L)}$ = 10, $t_{p(\overline{E})}$ = 20 ns, PRR = 1 MHz and $Z_{out} \approx 50\ \Omega$.



WAVEFORM C

NOTES:
1. The master reset inputs have the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq 10$ ns, $t_{p(\overline{MR})}$ = 20 ns and PRR = 1 MHz.
2. The enable inputs have the following characteristics: V_{gen} = 3 V, $t_0 = t_1 \leq = 10$ ns, t_{REC} = 10 ns, $t_{p(\overline{E})}$ = 20 ns, PRR = 1 MHz and $Z_{out} \approx 50\ \Omega$.

FIGURE 5. Switching time test circuit and waveforms for device type 02 - Continued.

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TABLE II. Group A inspection for device type 01.

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits		
			Test No.	E	$\overline{S}_0$	D_0	D_1	$\overline{S}_2$	D_2	D_3	GND	$\overline{MR}$	Q_3	$\overline{S}_3$	Q_2	Q_1	$\overline{S}_1$	Q_0	V_{CC}		Min	Max	Unit
1 T = 25°C	V_{OH}	3006	1 2 3 4	0.7 V	0.7 V	2.0 V	2.0 V	0.7 V	2.0 V	2.0 V	GND	2.0 V	-400 µA	0.8 V	-400 µA	-400 µA	0.7 V	-400 µA	4.5 V	Q_0 Q_1 Q_2 Q_3	2.4		V
	V_{OL}	3007	5 6 7 8			0.7 V	0.7 V		0.7 V	0.7 V			4.8 mA		4.8 mA	4.8 mA		4.8 mA		Q_0 Q_1 Q_2 Q_3		0.3	
	V_{IC}		9 10 11 12 13 14 15 16 17 18	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA		-10 mA		- 10 mA			-10 mA			$\overline{E}$ $\overline{S}_0$ D_0 D_1 $\overline{S}_2$ D_2 D_3 $\overline{MR}$ $\overline{S}_3$ S_1		-1.5	
	I_{IL1}	3009	19 20 21 22	GND	GND	0.3 V	0.3 V	GND	0.3 V	0.3 V				GND			GND		5.5 V	D_0 D_1 D_2 D_3	-282	-640	µA
	I_{IL2}		23 24 25 26 27 28	0.3 V	0.3 V			0.3 V				0.3 V		0.3 V			0.3 V			$\overline{E}$ S_0 $\overline{S}_2$ $\overline{MR}$ S_3 S_1	-176	-400	
	I_{IH1}	3010	29 30 31 32	5.5 V	5.5 V	2.4 V	2.4 V	5.5 V	2.4 V	2.4 V		GND					5.5 V			D_0 D_1 D_2 D_3		30	
	I_{IH2}		33 34 35 36			5.5 V	5.5 V		5.5 V	5.5 V										D_0 D_1 D_2 D_3		150	
	I_{IH3}		37 38 39 40 41 42	2.4 V	2.4 V			2.4 V				2.4 V		2.4 V			2.4 V			$\overline{E}$ $\overline{S}_0$ $\overline{S}_2$ $\overline{MR}$ $\overline{S}_3$ S_1		20	

See note at end of device type 01.

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TABLE III. Group A inspection for device type 01.[1] -Continued

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits Min	Max	Unit
1 T_C = 25°C	I_{IH4}	3010	43 44 45 46 47 48	5.5 V	5.5 V			5.5 V			GND	5.5 V		5.5 V			5.5 V		5.5 V	$\overline{E}$ $\overline{S}_0$ D_0 D_1 D_2 D_3 $\overline{MR}$ S_2 S_1		100	µA
	I_{OS}	3011	49 50 51 52	GND	GND	4.5 V	4.5 V	GND	4.5 V	4.5 V		5.5 V	GND	GND	GND	GND	GND	GND		Q_0 Q_1 Q_2 Q_3	-2.5	-25	mA
	I_{CC}	3005	53	4.5 V	GND			GND				GND		GND			GND			V_{CC}		16.5	
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = 25°C	Functional Tests	3014	54 55 56 57 58 59 60 61 62 63 64 65	L[2] L L L L H H L H H H L	L L H H H H L L L L L L	L H H L H H H H H H L H	L H H L H H H H H L L H	L L H H H H L L L L L L	L H H L H H H H H L L H	L H H L H H H H H L L H	GND	H[2] H H H H H H H H H L L	L H H L H L L L H H L L	L L H L H H H H L L L L	L H H L L L L L H H L L	L H H L L L L L H H L L	L L H H H H L L L L L L	L H H L L L L H H H L L	5.0 V		H or L as shown 3/		
8	Repeat subgroup 7 at T_C = 125°C and T_C = -55°C.																						
9 T_C = 25°C	t_{PLH1}	3003 (Fig. 5-A)	66 67 68 69	IN	GND	IN X X X	X IN X X	GND	X X IN X	X X X IN	GND	5.0 V	OUT	GND	OUT	OUT	GND	OUT	5.0 V	$\overline{E}$ $\overline{E}$ $\overline{E}$ $\overline{E}$	Q_0 Q_1 Q_2 Q_3	20	60	ns
	t_{PHL1}	3003 (Fig. 5-B)	70 71 72 73		GND	IN X X X	X IN X X	GND	X X IN X	X X X IN			OUT	GND	OUT	OUT	GND	OUT		$\overline{E}$ $\overline{E}$ $\overline{E}$ $\overline{E}$	Q_0 Q_1 Q_2 Q_3	15	52	
	t_{PHL1}	3003 (Fig 5-C)	74 75 76 77		GND	IN X X X	X IN X X	GND	X X IN X	X X X IN			OUT	GND	OUT	OUT	GND	OUT		$\overline{E}_0$ $\overline{E}_1$ $\overline{E}_2$ $\overline{E}_3$	Q_0 Q_1 Q_2 Q_3			

See note at end of device type 01.

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TABLE III. Group A inspection for device type 01. 1/ -Continued.

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1 $\overline{E}$	2 S_0	3 D_0	4 D_1	5 S_2	6 D_2	7 D_3	8 GND	9 $\overline{MR}$	10 Q_3	11 S_3	12 Q_2	13 Q_1	14 S_1	15 Q_0	16 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
9 T_C=25°C	t_{PLH2}	3003 (Fig. 5-B)	78	IN	GND	IN	X		X	X	GND	5.0 V						OUT	5.0 V	D_0 - Q_0	10	45	ns
			79			X	IN		X	X										D_1 - Q_1			
			80			X	X	GND	IN	X			OUT	GND	OUT	OUT	GND			D_2 - Q_2			
			81			X	X		X	IN										D_3 - Q_3			
	t_{PLH2}	3003 Fig. 5-C)	82		GND	IN	X		X	X								OUT		D_0 - Q_0			
			83			X	IN		X	X										D_1 - Q_1			
			84			X	X	GND	IN	X			OUT	GND	OUT	OUT	GND			D_2 - Q_2			
			85			X	X		X	IN										D_3 - Q_3			
	t_{PLH2}	3003 (Fig. 5-D)	86	GND	IN	IN	X	X	X	X				X			X	OUT		D_0 - Q_0			
			87		X	X	IN	X	X	X				X			IN			D_1 - Q_1			
			88		X	X	X	IN	IN	X			OUT	X	OUT	OUT	X			D_2 - Q_2			
			89		X	X	X	X	X	IN				IN			X			D_3 - Q_3			
	t_{PHL2}	3003 (Fig. 5-E)	90	GND	IN	IN	X	X	X	X				X			X	OUT		D_0 - Q_0			
			91		X	X	IN	X	X	X				X			IN			D_1 - Q_1			
			92		X	X	X	IN	IN	X			OUT	X	OUT	OUT	X			D_2 - Q_2			
			93		X	X	X	X	X	IN				IN			X			D_3 - Q_3			
	t_{PHL2}	3003 (Fig. 5-A)	94	IN	GND	IN	X		X	X								OUT		D_0 - Q_0			
			95			X	IN		X	X										D_1 - Q_1			
			96			X	X	GND	IN	X			OUT	GND	OUT	OUT	GND			D_2 - Q_2			
			97			X	X		X	IN										D_3 - Q_3			
	t_{PHL3}	3003 (Fig 5-F)	98		GND	5.0 V	5.0 V					IN						OUT		$\overline{MR}$ - Q_0			
			99					GND	5.0 V	5.0 V										$\overline{MR}$ - Q_1			
			100										OUT	GND	OUT	OUT	GND			$\overline{MR}$ - Q_2			
			101																	$\overline{MR}$ - Q_3			
	t_{PLH4}	3003 (Fig. 5-E)	102	GND	IN	IN	X	X	X	X		5.0 V		X			X	OUT		S_0 - Q_0	15	48	
			103		X	X	IN	X	X	X				X			IN			S_1 - Q_1			
			104		X	X	X	IN	IN	X			OUT	X	OUT	OUT	X			S_2 - Q_2			
			105		X	X	X	X	X	IN				IN			X			S_3 - Q_3			
10 T_C=125°C	t_{PLH1}	3003 (Fig. 5-A)	106	IN	GND	IN	X		X	X								OUT		E - Q_0	15	65	
			107			X	IN		X	X										E - Q_1			
			108			X	X	GND	IN	X			OUT	GND	OUT	OUT	GND			E - Q_2			
			109			X	X		X	IN										E - Q_3			
	t_{PHL1}	3003 (Fig. 5-B)	110		GND	IN	X		X	X								OUT		E - Q_0		57	
			111			X	IN		X	X										E - Q_1			
			112			X	X	GND	IN	X			OUT	GND	OUT	OUT	GND			E - Q_2			
			113			X	X		X	IN										E - Q_3			
	t_{PHL1}	3003 (Fig. 5-C)	114		GND	IN	X		X	X								OUT		E - Q_0			
			115			X	IN		X	X										E - Q_1			
			116			X	X	GND	IN	X			OUT	GND	OUT	OUT	GND			E - Q_2			
			117			X	X		X	IN										E - Q_3			

See notes at end of device type 01.

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TABLE III. Group A inspection for device type 01. 1/ -Continued

Subgroup	Symbol	MIL-STD-883 method	Case E, F Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Meas. terminal	Test limits Min	Max	Unit
				$\overline{E}$	$\overline{S}_0$	D_0	D_1	$\overline{S}_2$	D_2	D_3	GND	$\overline{MR}$	Q_3	$\overline{S}_3$	Q_2	Q_1	$\overline{S}_1$	Q_0	V_{CC}				
10 T_C = 125°C	t_{PLH2}	3003 (Fig. 5-B)	118	IN	GND	IN	X	GND	X	X	GND	5.0 V	OUT	GND	OUT	OUT	GND	OUT	5.0 V	D_0 - Q_0	8	50	ns
			119			X	IN		X	X										D_1 - Q_1			
			120			X	X		IN	X										D_2 - Q_2			
			121			X	X		X	IN										D_3 - Q_3			
	t_{PLH2}	3003 (Fig. 5-C)	122		GND	IN	X	GND	X	X			OUT	GND	OUT	OUT	GND	OUT		D_0 - Q_0			
			123			X	IN		X	X										D_1 - Q_1			
			124			X	X		IN	X										D_2 - Q_2			
			125			X	X		X	IN										D_3 - Q_3			
	t_{PLH2}	3003 (Fig. 5-D)	126	GND	GND	IN	X	X	X	X				X	OUT	OUT	X	OUT		D_0 - Q_0			
			127		X	X	IN	X	X	X				X			GND			D_1 - Q_1			
			128		X	X	X	GND	IN	X				X	OUT		X			D_2 - Q_2			
			129		X	X	X	X	X	IN			OUT	GND			X			D_3 - Q_3			
	t_{PHL2}	3003 (Fig. 5-E)	130	GND	GND	IN	X	X	X	X				X	OUT	OUT	X	OUT		D_0 - Q_0			
			131		X	X	IN	X	X	X				X			GND			D_1 - Q_1			
			132		X	X	X	GND	IN	X				X	OUT		X			D_2 - Q_2			
			133		X	X	X	X	X	IN			OUT	GND			X			D_3 - Q_3			
	t_{PHL2}	3003 (Fig. 5-A)	134	IN	GND	IN	X		X	X					OUT		OUT	OUT		D_0 - Q_0			
			135			X	IN		X	X								GND			D_1 - Q_1			
			136			X	X	GND	IN	X					OUT						D_2 - Q_2			
			137			X	X		X	IN			OUT	GND						D_3 - Q_3			
	t_{PHL3}	3003 (Fig. 5-F)	138		GND	5.0 V	5.0 V	GND	5.0 V	5.0 V		IN			OUT	OUT	GND	OUT		$\overline{MR}$ - Q_0			
			139																	$\overline{MR}$ - Q_1			
			140																	MR - Q_2			
			141										OUT	GND						$\overline{MR}$ - Q_3			
	t_{PLH4}	3003 (Fig. 5-E)	142	GND	IN	IN	X	X	X	X		5.0 V		X	OUT	OUT	X	OUT		S_0 - Q_0	12	53	
			143		X	X	IN	X	X	X				X			IN			S_1 - Q_1			
			144		X	X	X	IN	IN	X				X	OUT		X			S_2 - Q_2			
			145		X	X	X	X	X	IN			OUT	IN			X			S_3 - Q_3			
11	Same tests, terminal conditions and limits as subgroup 10, except T_C = -55°C.																						

NOTE:

X = Inputs may be low, high, or open.

1/ Terminal conditions (pins not designated may be H = 2.0 V or L = 0.7 V, or open).
2/ Input voltages shown are H = 2.0 V minimum and L = 0.7 V maximum.
3/ Output voltages shall be either: (a) H = 2.4 V minimum and L = 0.3 V maximum when using a high speed checker double comparator, or (b) H $\geq$ 1.5 V and L $<$ 1.5 V when using a high speed checker single comparator.

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TABLE III. Group A inspection for device type 08 1/

Subgroup	Symbol	MIL-STD-883 method	Cases J, K	Test No.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Meas. terminal	Test limits Min	Test limits Max	Unit
1 T_C=25°C	V_{OH}	3006			$\overline{1E}$	$1E_0$	$1E_1$	$1D_0$	$1Q_0$	$1D_1$	$1Q_1$	$1D_2$	$1Q_2$	$1D_3$	$1Q_3$	GND	$\overline{2E}$	$2E_0$	$2E_1$	$2D_0$	$2Q_0$	$2D_1$	$2Q_1$	$2D_2$	$2Q_2$	$2D_3$	$2Q_3$	V_{CC}			2.4		V
				1	4.5 V	0.7 V		4.5 V	-0.4 mA	4.5 V	-0.4 mA	4.5 V	-0.4 mA	4.5 V	-0.4 mA						-0.4 mA	4.5 V	-0.4 mA	4.5 V	-0.4 mA	4.5 V	-0.4 mA					
				2		0.7 V																										
				3			0.7 V																									
				4				4.5 V		4.5 V		4.5 V		4.5 V								4.5 V		4.5 V		4.5 V						
				5					-0.4 mA		-0.4 mA		-0.4 mA		-0.4 mA						-0.4 mA		-0.4 mA		-0.4 mA		-0.4 mA					
				6						4.5 V		4.5 V		4.5 V								4.5 V		4.5 V		4.5 V						
				7							-0.4 mA		-0.4 mA		-0.4 mA						-0.4 mA		-0.4 mA		-0.4 mA		-0.4 mA					
				8								4.5 V		4.5 V								4.5 V		4.5 V		4.5 V						
				9									-0.4 mA		-0.4 mA						-0.4 mA		-0.4 mA		-0.4 mA		-0.4 mA					
				10										4.5 V								4.5 V		4.5 V		4.5 V						
				11											-0.4 mA						-0.4 mA		-0.4 mA		-0.4 mA		-0.4 mA					
				12												GND																
				13													4.5 V															
				14														0.7 V														
				15															0.7 V													
				16																4.5 V												
				17					-0.4 mA												-0.4 mA											
				18						4.5 V												4.5 V										
				19							-0.4 mA												-0.4 mA									
				20								4.5 V												4.5 V								
				21									-0.4 mA												-0.4 mA							
				22										4.5 V												4.5 V						
				23											-0.4 mA												-0.4 mA					
				24	4.5 V																											
2 T_C=125°C	V_{OH}	3006			$\overline{1E}$	$1E_0$	$1E_1$	$1D_0$	$1Q_0$	$1D_1$	$1Q_1$	$1D_2$	$1Q_2$	$1D_3$	$1Q_3$	GND	$\overline{2E}$	$2E_0$	$2E_1$	$2D_0$	$2Q_0$	$2D_1$	$2Q_1$	$2D_2$	$2Q_2$	$2D_3$	$2Q_3$	V_{CC}			2.4		V
				1	4.5 V	0.7 V		4.5 V	-0.4 mA	4.5 V	-0.4 mA	4.5 V	-0.4 mA	4.5 V	-0.4 mA						-0.4 mA	4.5 V	-0.4 mA	4.5 V	-0.4 mA	4.5 V	-0.4 mA					
				2		0.7 V																										
				3			0.7 V																									
				4				4.5 V		4.5 V		4.5 V		4.5 V								4.5 V		4.5 V		4.5 V						
				5					-0.4 mA		-0.4 mA		-0.4 mA		-0.4 mA						-0.4 mA		-0.4 mA		-0.4 mA		-0.4 mA					
				6						4.5 V		4.5 V		4.5 V								4.5 V		4.5 V		4.5 V						
				7							-0.4 mA		-0.4 mA		-0.4 mA						-0.4 mA		-0.4 mA		-0.4 mA		-0.4 mA					
				8								4.5 V		4.5 V								4.5 V		4.5 V		4.5 V						
				9									-0.4 mA		-0.4 mA						-0.4 mA		-0.4 mA		-0.4 mA		-0.4 mA					
				10										4.5 V								4.5 V		4.5 V		4.5 V						
				11											-0.4 mA						-0.4 mA		-0.4 mA		-0.4 mA		-0.4 mA					
				12												GND																
				13													4.5 V															
				14														0.7 V														
				15															0.7 V													
				16																4.5 V												
				17					-0.4 mA												-0.4 mA											
				18						4.5 V												4.5 V										
				19							-0.4 mA												-0.4 mA									
				20								4.5 V												4.5 V								
				21									-0.4 mA												-0.4 mA							
				22										4.5 V												4.5 V						
				23											-0.4 mA												-0.4 mA					
				24	4.5 V																											
3 T_C=-55°C	V_{OH}	3006			$\overline{1E}$	$1E_0$	$1E_1$	$1D_0$	$1Q_0$	$1D_1$	$1Q_1$	$1D_2$	$1Q_2$	$1D_3$	$1Q_3$	GND	$\overline{2E}$	$2E_0$	$2E_1$	$2D_0$	$2Q_0$	$2D_1$	$2Q_1$	$2D_2$	$2Q_2$	$2D_3$	$2Q_3$	V_{CC}			2.4		V
				1	4.5 V	0.7 V		4.5 V	-0.4 mA	4.5 V	-0.4 mA	4.5 V	-0.4 mA	4.5 V	-0.4 mA						-0.4 mA	4.5 V	-0.4 mA	4.5 V	-0.4 mA	4.5 V	-0.4 mA					
				2		0.7 V																										
				3			0.7 V																									
				4				4.5 V		4.5 V		4.5 V		4.5 V								4.5 V		4.5 V		4.5 V						
				5					-0.4 mA		-0.4 mA		-0.4 mA		-0.4 mA						-0.4 mA		-0.4 mA		-0.4 mA		-0.4 mA					
				6						4.5 V		4.5 V		4.5 V								4.5 V		4.5 V		4.5 V						
				7							-0.4 mA		-0.4 mA		-0.4 mA						-0.4 mA		-0.4 mA		-0.4 mA		-0.4 mA					
				8								4.5 V		4.5 V								4.5 V		4.5 V		4.5 V						
				9									-0.4 mA		-0.4 mA						-0.4 mA		-0.4 mA		-0.4 mA		-0.4 mA					
				10										4.5 V								4.5 V		4.5 V		4.5 V						
				11											-0.4 mA						-0.4 mA		-0.4 mA		-0.4 mA		-0.4 mA					
				12												GND																
				13													4.5 V															
				14														0.7 V														
				15															0.7 V													
				16																4.5 V												
				17					-0.4 mA												-0.4 mA											
				18						4.5 V												4.5 V										
				19							-0.4 mA												-0.4 mA									
				20								4.5 V												4.5 V								
				21									-0.4 mA												-0.4 mA							
				22										4.5 V												4.5 V						
				23											-0.4 mA												-0.4 mA					
				24	4.5 V																											
1 T_C=25°C	V_{OL}	3007			$\overline{1E}$	$1E_0$	$1E_1$	$1D_0$	$1Q_0$	$1D_1$	$1Q_1$	$1D_2$	$1Q_2$	$1D_3$	$1Q_3$	GND	$\overline{2E}$	$2E_0$	$2E_1$	$2D_0$	$2Q_0$	$2D_1$	$2Q_1$	$2D_2$	$2Q_2$	$2D_3$	$2Q_3$	V_{CC}				0.3	V
				1	4.5 V	0.7 V		0.7 V	4.8 mA	0.7 V	4.8 mA	0.7 V	4.8 mA	0.7 V	4.8 mA						4.8 mA	0.7 V	4.8 mA	0.7 V	4.8 mA	0.7 V	4.8 mA					
				2		0.7 V																										
				3			0.7 V																									
				4				0.7 V		0.7 V		0.7 V		0.7 V								0.7 V		0.7 V		0.7 V						
				5					4.8 mA		4.8 mA		4.8 mA		4.8 mA						4.8 mA		4.8 mA		4.8 mA		4.8 mA					
				6						0.7 V		0.7 V		0.7 V								0.7 V		0.7 V		0.7 V						
				7							4.8 mA		4.8 mA		4.8 mA						4.8 mA		4.8 mA		4.8 mA		4.8 mA					
				8								0.7 V		0.7 V								0.7 V		0.7 V		0.7 V						
				9									4.8 mA		4.8 mA						4.8 mA		4.8 mA		4.8 mA		4.8 mA					
				10										0.7 V								0.7 V		0.7 V		0.7 V						
				11											4.8 mA						4.8 mA		4.8 mA		4.8 mA		4.8 mA					
				12												GND																
				13													4.5 V															
				14														0.7 V														
				15															0.7 V													
				16																4.5 V												
				17					4.8 mA												4.8 mA											
				18						0.7 V												0.7 V										
				19							4.8 mA												4.8 mA									
				20								0.7 V												0.7 V								
				21									4.8 mA												4.8 mA							
				22										0.7 V												0.7 V						
				23											4.8 mA												4.8 mA					
				24	4.5 V																											
2 T_C=125°C	V_{OL}	3007			$\overline{1E}$	$1E_0$	$1E_1$	$1D_0$	$1Q_0$	$1D_1$	$1Q_1$	$1D_2$	$1Q_2$	$1D_3$	$1Q_3$	GND	$\overline{2E}$	$2E_0$	$2E_1$	$2D_0$	$2Q_0$	$2D_1$	$2Q_1$	$2D_2$	$2Q_2$	$2D_3$	$2Q_3$	V_{CC}				0.4	V
				1	4.5 V	0.7 V		0.7 V	4.8 mA	0.7 V	4.8 mA	0.7 V	4.8 mA	0.7 V	4.8 mA						4.8 mA	0.7 V	4.8 mA	0.7 V	4.8 mA	0.7 V	4.8 mA					
				2		0.7 V																										
				3			0.7 V																									
				4				0.7 V		0.7 V		0.7 V		0.7 V								0.7 V		0.7 V		0.7 V						
				5					4.8 mA		4.8 mA		4.8 mA		4.8 mA						4.8 mA		4.8 mA		4.8 mA		4.8 mA					
				6						0.7 V		0.7 V		0.7 V								0.7 V		0.7 V		0.7 V						
				7							4.8 mA		4.8 mA		4.8 mA						4.8 mA		4.8 mA		4.8 mA		4.8 mA					
				8								0.7 V		0.7 V								0.7 V		0.7 V		0.7 V						
				9									4.8 mA		4.8 mA						4.8 mA		4.8 mA		4.8 mA		4.8 mA					
				10										0.7 V								0.7 V		0.7 V		0.7 V						
				11											4.8 mA						4.8 mA		4.8 mA		4.8 mA		4.8 mA					
				12												GND																
				13													4.5 V															
				14														0.7 V														
				15															0.7 V													
				16																4.5 V												
				17					4.8 mA												4.8 mA											
				18						0.7 V												0.7 V										
				19							4.8 mA												4.8 mA									
				20								0.7 V												0.7 V								
				21									4.8 mA												4.8 mA							
				22										0.7 V												0.7 V						
				23											4.8 mA												4.8 mA					
				24	4.5 V																											
3 T_C=-55°C	V_{OL}	3007			$\overline{1E}$	$1E_0$	$1E_1$	$1D_0$	$1Q_0$	$1D_1$	$1Q_1$	$1D_2$	$1Q_2$	$1D_3$	$1Q_3$	GND	$\overline{2E}$	$2E_0$	$2E_1$	$2D_0$	$2Q_0$	$2D_1$	$2Q_1$	$2D_2$	$2Q_2$	$2D_3$	$2Q_3$	V_{CC}				0.3	V
				1	4.5 V	0.7 V		0.7 V	4.8 mA	0.7 V	4.8 mA	0.7 V	4.8 mA	0.7 V	4.8 mA						4.8 mA	0.7 V	4.8 mA	0.7 V	4.8 mA	0.7 V	4.8 mA					
				2		0.7 V																										
				3			0.7 V																									
				4				0.7 V		0.7 V		0.7 V		0.7 V								0.7 V		0.7 V		0.7 V						
				5					4.8 mA		4.8 mA		4.8 mA		4.8 mA						4.8 mA		4.8 mA		4.8 mA		4.8 mA					
				6						0.7 V		0.7 V		0.7 V								0.7 V		0.7 V		0.7 V						
				7							4.8 mA		4.8 mA		4.8 mA						4.8 mA		4.8 mA		4.8 mA		4.8 mA					
				8								0.7 V		0.7 V								0.7 V		0.7 V		0.7 V						
				9									4.8 mA		4.8 mA						4.8 mA		4.8 mA		4.8 mA		4.8 mA					
				10										0.7 V								0.7 V		0.7 V		0.7 V						
				11											4.8 mA						4.8 mA		4.8 mA		4.8 mA		4.8 mA					
				12												GND																
				13													4.5 V															
				14														0.7 V														
				15															0.7 V													
				16																4.5 V												
				17					4.8 mA												4.8 mA											
				18						0.7 V												0.7 V										
				19							4.8 mA												4.8 mA									
				20								0.7 V												0.7 V								
				21									4.8 mA												4.8 mA							
				22										0.7 V												0.7 V						
				23											4.8 mA												4.8 mA					
				24	4.5 V																											
1 T_C=25°C	V_{IC}				$\overline{1E}$	$1E_0$	$1E_1$	$1D_0$	$1Q_0$	$1D_1$	$1Q_1$	$1D_2$	$1Q_2$	$1D_3$	$1Q_3$	GND	$\overline{2E}$	$2E_0$	$2E_1$	$2D_0$	$2Q_0$	$2D_1$	$2Q_1$	$2D_2$	$2Q_2$	$2D_3$	$2Q_3$	V_{CC}					V
				17			-10 mA	-10 mA		-10 mA		-10 mA		-10 mA																		
				18																		-10 mA										
				19																												
				20																				-10 mA								
				21																												
				22																						-10 mA						
				23																												
				24																												
2 T_C=125°C	V_{IC}				$\overline{1E}$	$1E_0$	$1E_1$	$1D_0$	$1Q_0$	$1D_1$	$1Q_1$	$1D_2$	$1Q_2$	$1D_3$	$1Q_3$	GND	$\overline{2E}$	$2E_0$	$2E_1$	$2D_0$	$2Q_0$	$2D_1$	$2Q_1$	$2D_2$	$2Q_2$	$2D_3$	$2Q_3$	V_{CC}					V
				17			-10 mA	-10 mA		-10 mA		-10 mA		-10 mA																		
				18																		-10 mA										
				19																												
				20																				-10 mA								
				21																												
				22																						-10 mA						
				23																												
				24																												
3 T_C=-55°C	V_{IC}				$\overline{1E}$	$1E_0$	$1E_1$	$1D_0$	$1Q_0$	$1D_1$	$1Q_1$	$1D_2$	$1Q_2$	$1D_3$	$1Q_3$	GND	$\overline{2E}$	$2E_0$	$2E_1$	$2D_0$	$2Q_0$	$2D_1$	$2Q_1$	$2D_2$	$2Q_2$	$2D_3$	$2Q_3$	V_{CC}					V
				17			-10 mA	-10 mA		-10 mA		-10 mA		-10 mA																		
				18																		-10 mA										
				19																												
				20																				-10 mA								
				21																												
				22																						-10 mA						
				23																												
				24																												
1 T_C=25°C	I_{IL1}	3009			$\overline{1E}$	$1E_0$	$1E_1$	$1D_0$	$1Q_0$	$1D_1$	$1Q_1$	$1D_2$	$1Q_2$	$1D_3$	$1Q_3$	GND	$\overline{2E}$	$2E_0$	$2E_1$	$2D_0$	$2Q_0$	$2D_1$	$2Q_1$	$2D_2$	$2Q_2$	$2D_3$	$2Q_3$	V_{CC}			-182	-640	μA
				31			GND									GND																
				32																												
				33																												
				34																												
				35																												
				36																												
				37																												
				38																												
				39	0.3 V																											
				40			0.3 V																									
				41			5.5 V																									
				42			0.3 V																									
				43																												
				44																												
2 T_C=125°C	I_{IL1}	3009			$\overline{1E}$	$1E_0$	$1E_1$	$1D_0$	$1Q_0$	$1D_1$	$1Q_1$	$1D_2$	$1Q_2$	$1D_3$	$1Q_3$	GND	$\overline{2E}$	$2E_0$	$2E_1$	$2D_0$	$2Q_0$	$2D_1$	$2Q_1$	$2D_2$	$2Q_2$	$2D_3$	$2Q_3$	V_{CC}			-182	-640	μA
				31			GND									GND																
				32																												
				33																												
				34																												
				35																												
				36																												
				37																												
				38																												
				39	0.3 V																											
				40			0.3 V																									
				41			5.5 V																									
				42			0.3 V																									
				43																												
				44																												
3 T_C=-55°C	I_{IL1}	3009			$\overline{1E}$	$1E_0$	$1E_1$	$1D_0$	$1Q_0$	$1D_1$	$1Q_1$	$1D_2$	$1Q_2$	$1D_3$	$1Q_3$	GND	$\overline{2E}$	$2E_0$	$2E_1$	$2D_0$	$2Q_0$	$2D_1$	$2Q_1$	$2D_2$	$2Q_2$	$2D_3$	$2Q_3$	V_{CC}			-182	-640	μA
				31			GND									GND																
				32																												
				33																												
				34																												
				35																												
				36																												
				37																												
				38																												
				39	0.3 V																											
				40			0.3 V																									
				41			5.5 V																									
				42			0.3 V																									
				43																												
				44																												
1 T_C=25°C	I_{IL2}	3009			$\overline{1E}$	$1E_0$	$1E_1$	$1D_0$	$1Q_0$	$1D_1$	$1Q_1$	$1D_2$	$1Q_2$	$1D_3$	$1Q_3$	GND	$\overline{2E}$	$2E_0$	$2E_1$	$2D_0$	$2Q_0$	$2D_1$	$2Q_1$	$2D_2$	$2Q_2$	$2D_3$	$2Q_3$	V_{CC}			-176	-600	μA
				31			GND									GND																
				32																												
				33																												
				34																												
				35																												
				36																												
				37																												
				38																												
				39	0.3 V																											
				40			0.3 V																									
				41			5.5 V																									
				42			0.3 V																									
				43																												
				44																												
2 T_C=125°C	I_{IL2}	3009			$\overline{1E}$	$1E_0$	$1E_1$	$1D_0$	$1Q_0$	$1D_1$	$1Q_1$	$1D_2$	$1Q_2$	$1D_3$	$1Q_3$	GND	$\overline{2E}$	$2E_0$	$2E_1$	$2D_0$	$2Q_0$	$2D_1$	$2Q_1$	$2D_2$	$2Q_2$	$2D_3$	$2Q_3$	V_{CC}			-176	-600	μA
				31			GND									GND																
				32																												
				33																												
				34																												
				35														

TABLE II. Group A inspection for device type 02. 1/ -Continued

Subgroup	Symbol	MIL-STD-883 method	Cases J, K Test No.	1 $1\overline{MR}$	2 $1\overline{E}_0$	3 $1\overline{E}_1$	4 $1D_0$	5 $1Q_0$	6 $1D_1$	7 $1Q_1$	8 $1D_2$	9 $1Q_2$	10 $1D_3$	11 $1Q_3$	12 GND	13 $2\overline{MR}$	14 $2\overline{E}_0$	15 $2\overline{E}_1$	16 $2D_0$	17 $2Q_0$	18 $2D_1$	19 $2Q_1$	20 $2D_2$	21 $2Q_2$	22 $2D_3$	23 $2Q_3$	24 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
1 T_C = 25°C	I_{IH1}	3010	45 46 47 48 49 50 51 52			5.5 V	2.4 V		2.4 V		2.4 V		2.4 V		GND			5.5 V	2.4 V		2.4 V		2.4 V		2.4 V		5.5 V	$1D_0$ $1D_1$ $1D_2$ $1D_3$ $2D_0$ $2D_1$ $2D_2$ $2D_3$		30	µA
	I_{IH2}	3010	53 54 55 56 57 58 59 60			5.5 V	5.5 V		5.5 V		5.5 V		5.5 V					5.5 V	5.5 V		5.5 V		5.5 V		5.5 V			$1D_0$ $1D_1$ $1D_2$ $1D_3$ $2D_0$ $2D_1$ $2D_2$ $2D_3$		150		
	I_{IH3}	3010	61 62 63 64 65 66	2.4 V	2.4 V GND	GND 2.4 V										2.4 V	2.4 V GND	GND 2.4 V										$1\overline{MR}$ $1\overline{E}_0$ $1\overline{E}_1$ $2\overline{MR}$ $2\overline{E}_0$ $2\overline{E}_1$		20		
	I_{IH4}		67 68 69 70 71 72	5.5 V	5.5 V GND	GND 5.5 V										5.5 V	5.5 V GND	GND 5.5 V										$1\overline{MR}$ $1\overline{E}_0$ $1\overline{E}_1$ $2\overline{MR}$ $2\overline{E}_0$ $2\overline{E}_1$		100		
	I_{OS}	3011	73 74 75 76 77 78 79 80	4.5 V	GND	GND	4.5 V	GND	4.5 V	GND	4.5 V	GND	4.5 V	GND		4.5 V	GND	GND	4.5 V	GND	4.5 V	GND	4.5 V	GND	4.5 V	GND		$1Q_0$ $1Q_1$ $1Q_2$ $1Q_3$ $2Q_0$ $2Q_1$ $2Q_2$ $2Q_3$	-2.5	-25	mA	
	I_{CC}	3005	81	4.5 V	4.5 V	4.5 V	4.5 V		4.5 V		4.5 V		4.5 V			4.5 V	4.5 V	4.5 V	4.5 V		4.5 V		4.5 V		4.5 V			V_{CC}		-29		
2	Same tests, terminal conditions and limits as for subgroup 1, except T_C = 125°C and V_{IC} tests are omitted.																													
3	Same tests, terminal conditions and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																													

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TABLE III. Group A inspection for device type 02. 1/ -Continued

Subgroup	Symbol	MIL-STD-883 method	Case J, K Test No.	1 $1\overline{MR}$	2 $1E_0$	3 $1\overline{E}_1$	4 $1D_0$	5 $1Q_0$	6 $1D_1$	7 $1Q_1$	8 $1D_2$	9 $1Q_2$	10 $1D_3$	11 $1Q_3$	12 GND	13 $2\overline{MR}$	14 $2\overline{E}_0$	15 $2\overline{E}_1$	16 $2D_0$	17 $2Q_0$	18 $2D_1$	19 $2Q_1$	20 $2D_2$	21 $2Q_2$	22 $2D_3$	23 $2Q_3$	24 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
7 T_C = 25°C	Functional tests	3014	82	L 2/	H 2/	H	H	L	H	L	H	L	H	L	GND	L	H	H	L	H	L	H	L	H	L	5.0 V		H or L as shown 3/		
			83	H	H	H	H	L	H	L	H	L	H	L		H	H	H	H	L	H	L	H	L	H	L					
			84	H	L	L	H	H	H	H	H	H	H	H		H	L	L	H	H	R	H	H	H	H	H					
			85	H	L	L	L	L	H	H	H	H	H	H		H	L	L	L	L	H	H	H	H	H	H					
			86	H	L	L	L	L	L	L	H	H	H	H		H	L	L	L	L	L	L	H	R	H	H					
			87	H	L	L	L	L	L	L	L	L	H	H		H	L	L	L	L	L	L	L	L	H	H					
			88	H	L	L	L	L	L	L	L	L	L	L		H	L	L	L	L	L	L	L	L	L	L					
			89	H	L	L	H	H	L	L	L	L	L	L		H	L	L	H	H	L	L	L	L	L	L					
			90	H	L	L	H	H	H	H	L	L	L	L		H	L	L	H	H	H	H	H	L	L	L					
			91	H	L	L	H	H	H	H	H	H	L	L		H	L	L	H	H	H	H	H	H	H	L					
			92	H	L	L	H	H	H	H	H	H	H	H		H	L	L	H	H	H	H	H	H	H	H					
			93	H	H	L	H	H	H	H	H	H	H	H		H	H	L	H	H	H	H	H	H	H	H					
			94	H	H	L	L	L	H	L	L	H	L	H		H	H	L	H	L	L	H	L	H	L	H					
			95	H	H	H	L	H	H	L	L	H	L	H		H	H	L	H	L	L	H	L	H	L	H					
			96	H	L	H	L	H	H	L	L	H	L	H		H	L	L	H	L	L	H	L	H	L	H					
			97	L	L	L	H	L	H	L	H	L	H	L		L	L	L	H	L	R	L	R	L	H	L					
			98	L	H	L	H	L	H	L	H	L	H	L		L	H	L	H	L	H	L	H	L	H	L					
			99	H	H	L	H	L	H	L	H	L	H	L		H	H	L	H	L	H	L	H	L	H	L					
			100	H	H	H	H	L	H	L	H	L	H	L		H	H	L	H	L	H	L	H	L	H	L					
			101	H	L	H	H	L	H	L	H	L	H	L		H	L	H	H	L	H	L	H	L	H	L					
8	Repeat subgroup 7 at 125°C and -55°C.																													
9 T_C = 25°C	t_{PLH1}	3003 (Fig. 6-B)	102	5.0 V	IN	GND	IN	OUT	X	OUT	X	OUT	X	OUT	GND	5.0 V	IN	GND	IN	OUT	X	OUT	X	OUT	X	OUT	5.0 V	$1E_0$ - $1Q_0$	24	60	ns
			103				X		IN		X		X						X		IN		X		X			$1E_1$ - $1Q_1$			
			104				X		X		IN		X						X		X		IN		X			$1E_2$ - $1Q_2$			
			105				X		X		X		IN						X		X		X		IN			$1E_3$ - $1Q_3$			
			106		GND		IN	OUT	X		X		X						IN		X		X		X			$2E_0$ - $2Q_0$			
			107				X		X	OUT	X		X						X		X		X		X			$2E_1$ - $2Q_1$			
			108				X		IN		X	OUT	X						X		X		X		X			$2E_2$ - $2Q_2$			
			109						X		IN		X						X		X		X		X			$2E_3$ - $2Q_3$			
			110								X		IN						X		X		X		X						
			111										X						X		X		X		X						
			112										X						X		X		X		X						
			113										X						X		X		X		X						
			114										X						X		X		X		X						
			115										X						X		X		X		X						
			116										X						X		X		X		X						
			117										X						X		X		X		X						
	t_{PHL1}	3003 (Fig. 6-A)	118		IN	GND	IN	OUT	X	OUT	X	OUT	X	OUT														$1E_0$ - $1Q_0$			
			119				X		IN		X		X															$1E_1$ - $1Q_1$			
			120				X		X		IN	OUT	X															$1E_2$ - $1Q_2$			
			121				X		X		X		IN															$1E_3$ - $1Q_3$			
			122		GND		IN	OUT	X		X		X	OUT														$2E_0$ - $2Q_0$			
			123				X		X	OUT	X		X															$2E_1$ - $2Q_1$			
			124				X		IN		X	OUT	X															$2E_2$ - $2Q_2$			
			125				X		X		IN		X	OUT														$2E_3$ - $2Q_3$			

See notes at end of device type 02.

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TABLE III Group A inspection for device type 02. 1/ -Continued

Subgroup	Symbol	MIL-STD-883 method	Case J K Test No.	1 $1\overline{MR}$	2 $1\overline{E}_0$	3 $1\overline{E}_1$	4 $1D_0$	5 $1Q_0$	6 $1D_1$	7 $1Q_1$	8 $1D_2$	9 $1Q_2$	10 $1D_3$	11 $1Q_3$	12 GND	13 $2\overline{MR}$	14 $2\overline{E}_0$	15 $2\overline{E}_1$	16 $2D_0$	17 $2Q_0$	18 $2D_1$	19 $2Q_1$	20 $2D_2$	21 $2Q_2$	22 $2D_3$	23 $2Q_3$	24 V_{CC}	Meas. terminal	Test limits Min	Max	Unit
9 T_C=25°C	t_{PHL1}	3003 (Fig. 6-B)	126	5.0 V											GND	5.0 V	IN	GND	IN	OUT	X	X	X	X	5.0 V	$2\overline{E}_0$-$2Q_0$	24	60	ns
			127																X		IN	OUT	X			$2\overline{E}_1$-$2Q_1$			
			128																X		X		IN	OUT		$2\overline{E}_2$-$2Q_2$			
			129																X		X		X		OUT	$2\overline{E}_3$-$2Q_3$			
			130														GND	IN	IN	OUT	X		X		X		$2\overline{E}_0$-$2Q_0$			
			131																X		IN	OUT	X		X		$2\overline{E}_1$-$2Q_1$			
			132																X		X		IN	OUT	X		$2\overline{E}_2$-$2Q_2$			
			133																X		X		X		IN	OUT		$2\overline{E}_3$-$2Q_3$			
	t_{PHL2}		134		IN	GND	IN	OUT	X		X		X															$1D_0$-$1Q_0$	16	36	
			135				X		IN	OUT	X		X															$1D_1$-$1Q_1$			
			136				X		X		IN	OUT	X															$1D_2$-$1Q_2$			
			137				X		X		X		IN	OUT														$1D_3$-$1Q_3$			
			138		GND	IN	IN	OUT	X		X		X															$1D_0$-$1Q_0$			
			139				X		IN	OUT	X		X															$1D_1$-$1Q_1$			
			140				X		X		IN	OUT	X															$1D_2$-$1Q_2$			
			141				X		X		X		IN	OUT														$1D_3$-$1Q_3$			
			142														IN	GND	IN	OUT	X		X		X			$2D_0$-$2Q_0$			
			143																X		IN	OUT	X		X			$2D_1$-$2Q_1$			
			144																X		X		IN	OUT	X			$2D_2$-$2Q_2$			
			145																X		X		X		IN	OUT		$2D_3$-$2Q_3$			
			146														GND	IN	IN	OUT	X		X		X			$2D_0$-$2Q_0$			
			147																X		IN	OUT	X		X			$2D_1$-$2Q_1$			
			148																X		X		IN	OUT	X			$2D_2$-$2Q_2$			
			149																X		X		X		IN	OUT		$2D_3$-$2Q_3$			
	t_{PHL2}	3003 (Fig. 6)	150		IN	GND	IN	OUT	X		X		X															$1D_0$-$1Q_0$	22	60	
			151				X		IN	OUT	X		X															$1D_1$-$1Q_1$			
			152				X		X		IN	OUT	X															$1D_2$-$1Q_2$			
			153				X		X		X		IN	OUT														$1D_3$-$1Q_3$			
			154		GND	IN	IN	OUT	X		X		X															$1D_0$-$1Q_0$			
			155				X		IN	OUT	X		X															$1D_1$-$1Q_1$			
			156				X		X		IN	OUT	X															$1D_2$-$1Q_2$			
			157				X		X		X		IN	OUT														$1D_3$-$1Q_3$			
			158														IN	GND	IN	OUT	X		X		X			$2D_0$-$2Q_0$			
			159																X		IN	OUT	X		X			$2D_1$-$2Q_1$			
			160																X		X		IN	OUT	X			$2D_2$-$2Q_2$			
			161																X		X		X		IN	OUT		$2D_3$-$2Q_3$			
			162														GND	IN	IN	OUT	X		X		X			$2D_0$-$2Q_0$			
			163																X		IN	OUT	X		X			$2D_1$-$2Q_1$			
			164																X		X		IN	OUT	X			$2D_2$-$2Q_2$			
			165																X		X		X		IN	OUT		$2D_3$-$2Q_3$			
	t_{PHL3}	3003 (Fig. 6-C)	166	IN	IN	GND	5.0 V	OUT	5.0 V	OUT	5.0 V	OUT	5.0 V	OUT														$1\overline{MR}$-$1Q_0$	18	48	
			167																									$1\overline{MR}$-$1Q_1$			
			168																									$1\overline{MR}$-$1Q_2$			
			169																									$1\overline{MR}$-$1Q_3$			
			170		GND	IN		OUT																				$1\overline{MR}$-$1Q_0$			
			171							OUT																		$1\overline{MR}$-$1Q_1$			
			172									OUT																$1\overline{MR}$-$1Q_2$			

See notes at end of device type 02

MIL-M-38510/64B

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TABLE III. Group A inspection for device type 02. 1/ -Continued

Subgroup	Symbol	MIL-STD-883 method	Cases J, K Test No	1 $1\overline{MR}$	2 $1E_0$	3 $1E_1$	4 $1D_0$	5 $1Q_0$	6 $1D_1$	7 $1Q_1$	8 $1D_2$	9 $1Q_2$	10 $1D_3$	11 $1Q_3$	12 GND	13 $2\overline{MR}$	14 $2E_0$	15 $2E_1$	16 $2D_0$	17 $2Q_0$	18 $2D_1$	19 $2Q_1$	20 $2D_2$	21 $2Q_2$	22 $2D_3$	23 $2Q_3$	24 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Test limits Unit
9 T_C=25°C	t_{PHL2}	3003 (Fig. 6)	173	IN	GND	IN	5.0 V		5.0 V		5.0 V		5.0 V		GND	IN	IN → GND	GND → IN	5.0 V	OUT	5.0 V	OUT	5.0 V	OUT	5.0 V	OUT	5.0 V	1MR : 1Q1			
			174																									1MR : 2Q0			
			175																									1MR : 2Q1			
			176																									1MR : 2Q2			
			177																									1MR : 2Q3			
			178																									2MR : 1Q1			
			179																									2MR : 1Q2			
			180																									2MR : 1Q3			
			181																									2MR : 2Q1			
10 T_C=125°C	t_{PLH1}	3003 (Fig. 6-B)	182	5.0 V	IN → GND	GND → IN	X	OUT	X	OUT	X	OUT	X	OUT	GND	5.0 V	IN → GND	GND → IN	X	OUT	X	OUT	X	OUT	X	OUT	5.0 V	1E0 : 1Q1	18	80	ns
			183				IN		IN		IN		IN						IN		IN		IN		IN			1E1 : 1Q1			
			184				X		X		X		X						X		X		X		X			1E0 : 1Q0			
			185				X		X		X		X						X		X		X		X			1E1 : 1Q0			
			186				X		X		X		X						X		X		X		X			1E0 : 1Q1			
			187				X		X		X		X						X		X		X		X			1E1 : 1Q1			
			188				X		X		X		X						X		X		X		X			1E0 : 1Q2			
			189				X		X		X		X						X		X		X		X			1E1 : 1Q2			
			190				X		X		X		X						X		X		X		X			1E0 : 1Q3			
			191				X		X		X		X						X		X		X		X			1E1 : 1Q3			
			192				X		X		X		X						X		X		X		X			1E0 : 2Q0			
			193				X		X		X		X						X		X		X		X			1E1 : 2Q0			
			194				X		X		X		X						X		X		X		X			1E0 : 2Q1			
			195				X		X		X		X						X		X		X		X			1E1 : 2Q1			
			196				X		X		X		X						X		X		X		X			1E0 : 2Q2			
			197				X		X		X		X						X		X		X		X			1E1 : 2Q2			
	t_{PHL1}	3003 (Fig. 6-A)	198	5.0 V	IN → GND	GND → IN	X	OUT	X	OUT	X	OUT	X	OUT	GND	5.0 V	IN → GND	GND → IN	X	OUT	X	OUT	X	OUT	X	OUT	5.0 V	1E0 : 1Q1	20	80	ns
			199				IN		IN		IN		IN						IN		IN		IN		IN			1E1 : 1Q1			
			200				X		X		X		X						X		X		X		X			1E0 : 1Q0			
			201				X		X		X		X						X		X		X		X			1E1 : 1Q0			
			202				X		X		X		X						X		X		X		X			1E0 : 1Q1			
			203				X		X		X		X						X		X		X		X			1E1 : 1Q1			
			204				X		X		X		X						X		X		X		X			1E0 : 1Q2			
			205				X		X		X		X						X		X		X		X			1E1 : 1Q2			
			206				X		X		X		X						X		X		X		X			1E0 : 1Q3			
			207				X		X		X		X						X		X		X		X			1E1 : 1Q3			
			208				X		X		X		X						X		X		X		X			1E0 : 2Q0			
			209				X		X		X		X						X		X		X		X			1E1 : 2Q0			
			210				X		X		X		X						X		X		X		X			1E0 : 2Q1			
			211				X		X		X		X						X		X		X		X			1E1 : 2Q1			
			212				X		X		X		X						X		X		X		X			1E0 : 2Q2			
			213				X		X		X		X						X		X		X		X			1E1 : 2Q2			
	t_{PLH2}	3003 (Fig. 6)	214	IN	IN → GND	GND → IN	IN	OUT	IN	OUT	IN	OUT	IN	OUT	GND	IN	IN → GND	GND → IN	IN	OUT	IN	OUT	IN	OUT	IN	OUT	5.0 V	1D0 : 1Q1	15	43	ns
			215																									1D1 : 1Q1			
			216																									1D2 : 1Q1			
			217																									1D3 : 1Q1			
			218																									1D0 : 1Q2			
			219																									1D1 : 1Q2			
			220																									1D2 : 1Q2			
			221																									1D3 : 1Q2			

See notes at end of device type 02.

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TABLE III. Group A inspection for device type 02 1/ - Continued

Subgroup	Symbol	MIL-STD-883 method	Class J, K Test No.	1 $\overline{ERR}$	2 IE_0	3 IE_1	4 ID_0	5 IQ_0	6 ID_1	7 IQ_1	8 ID_2	9 IQ_2	10 ID_3	11 IQ_3	12 GND	13 $\overline{ERR}$	14 IE_0	15 IE_1	16 ID_0	17 IQ_0	18 ID_1	19 IQ_1	20 ID_2	21 IQ_2	22 ID_3	23 IQ_3	24 V_{CC}	Meas. terminal	Test Inputs: Max Clock	Min Clock
10 T_C=125°C	$^tPLH2$	3003 (Fig. 6)	232 233 234 235 236 237 238	5.0 V											GND	5.0 V	IN	GND									5.0 V				
			239 240 241 242 243 244 245																	OUT		OUT		IN	OUT	OUT				
	$^tPHL2$		246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261		IN	GND	IN	OUT	IN	OUT	IN	OUT	IN	OUT			IN	GND	IN	OUT	IN	OUT	IN	OUT	5.0 V	OUT					
11	Same tests, terminal conditions and limits as for subgroup 10, except T_C = -55°C.																													

NOTES:
X - Inputs may be high, low, or open.
1/ Terminal conditions (pins not designated may be H = 2.0 V or L = 0.7 V, or open.
2/ Input voltages above are H = 2.0 V minimum and L = 0.7 V maximum.
3/ Output voltages shall be either:
a. H = 2.4 V minimum and L = 0.5 V maximum when using a high speed checker double comparator, or
b. H ≥ 1.5 V and L < 1.5 V when using a high speed checker single comparator.

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4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical test parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging of microcircuits shall be as specified in MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for packaging and packing.

f. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - Electrical ground (common terminal)

I_{IN} - - - - - - - - - - - Current flowing into an input terminal

V_{IN} - - - - - - - - - - - Voltage level at an input terminal

T_C - - - - - - - - - - - - Case temperature

6.5 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

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6.6 Generic test data. Generic test data may be used to satisfy the requirements of 4.4.3. Group C generic test data shall be on date codes no more than one year old and on a die in the same microcircuit group (see appendix E of MIL-M-38510) with the same material, design and process and from the same plant as the die represented. Group D (see 4.4.3) generic data shall be on date codes no more than one year old and on the same package type (see terms, definitions, and symbols of MIL-M-38510) and from the same plant as the package represented. The vendor is required to retain the generic data for a period of not less than 36 months from the date of shipment.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	93L14
02	93L08

6.8 Ordering guidance. Since the qualification requirements have been removed from the specification, orders may be placed immediately.

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/45B	MICROCIRCUITS, DIGITAL, TTL, LOW POWER, BISTABLE LATCHES, MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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